

CYBERSECURITY & ETHICAL HACKING

INTERNSHIP PROGRAM

ApexPlanet Software Pvt. Ltd.

TASK 1: FOUNDATION & ENVIRONMENT SETUP

Lab Setup Report

Timeline: Days 1 - 12

Report Type	Lab Setup Report — Task 1 Submission
Program	Cybersecurity & Ethical Hacking Internship (60 Days)
Organization	ApexPlanet Software Pvt. Ltd.
Task Duration	Days 1–12
Submitted By	Diwakar Chandra
Offer Letter ID	APSPL2632803
Date	6 May 2026

1. Executive Summary

This report documents the successful completion of Task 1 of the ApexPlanet Cybersecurity & Ethical Hacking Internship Program. Over Days 1 through 12, foundational knowledge in cybersecurity was established, a professional ethical hacking lab environment was configured, and core tools were familiarized.

The key activities completed include: setting up a virtualized attack-defense lab using Kali Linux and Metasploitable2, studying core cybersecurity concepts (CIA Triad, threat types, attack vectors), building Linux command-line proficiency, understanding networking fundamentals, exploring cryptography principles, and getting hands-on with essential security tools including Wireshark, Nmap, Burp Suite, and Netcat.

2. Task Objective

Build strong fundamentals in cybersecurity, networking, and cryptography, and set up a professional ethical hacking lab environment that will serve as the testing ground for all subsequent tasks in the internship program.

3. Cybersecurity Basics

3.1 The CIA Triad

The CIA Triad is the foundational model of information security, representing three core principles:

- **Confidentiality:** Ensures information is accessible only to those authorized to access it. Techniques include encryption, access control lists (ACLs), and multi-factor authentication (MFA). Example: Encrypting sensitive files using AES-256 prevents unauthorized access even if the storage media is stolen.
- **Integrity:** Guarantees that data is accurate and has not been tampered with. Techniques include hashing (SHA-256, MD5), digital signatures, and checksums. Example: Using SHA-256 to verify a downloaded file ensures the file was not modified in transit.
- **Availability:** Ensures that systems and data are accessible to authorized users when needed. Techniques include redundancy, failover systems, DDoS protection, and regular backups. Example: Load balancers ensure web services remain available during high traffic or partial server failure.

3.2 Threat Types Studied

The following common cyber threats were explored in depth:

- **Phishing:** Social engineering attack via deceptive emails/websites to steal credentials. Mitigation: Security awareness training, email filtering, SPF/DKIM records.
- **Malware:** Malicious software including viruses, trojans, worms, and spyware. Mitigation: Antivirus software, sandboxing, and application whitelisting.
- **DDoS (Distributed Denial of Service):** Overwhelming a target with traffic from multiple sources. Mitigation: Rate limiting, CDN-based filtering, traffic scrubbing.

- SQL Injection: Inserting malicious SQL code into input fields to manipulate databases. Mitigation: Prepared statements, parameterized queries, input validation.
- Brute Force: Systematically trying all possible passwords until the correct one is found. Mitigation: Account lockout policies, CAPTCHA, MFA.
- Ransomware: Encrypts victim data and demands ransom for the decryption key. Mitigation: Offline backups, network segmentation, patch management.

3.3 Attack Vectors

- Social Engineering: Manipulating individuals psychologically to divulge confidential information or perform actions that compromise security.
- Wireless Attacks: Exploiting weaknesses in Wi-Fi protocols (WEP/WPA vulnerabilities), rogue access points, and evil twin attacks.
- Insider Threats: Malicious or negligent actions by employees, contractors, or trusted users who have authorized access to systems.

4. Lab Environment Setup

A fully isolated virtual lab was configured to practice offensive and defensive cybersecurity techniques safely and legally. All activities were performed in a closed network with no external exposure.

4.1 Virtualization Platform

- Hypervisor: VirtualBox (open-source, cross-platform) was installed as the virtualization platform.
- Network Configuration: Host-Only Adapter was configured to create a completely isolated private network between virtual machines, preventing any traffic from reaching the internet.

4.2 Virtual Machines Configured

Component	Details	Status
Kali Linux 2024.x	Attacker machine — primary pentesting OS	Completed
Metasploitable2	Intentionally vulnerable target VM	Completed
DVWA (Damn Vulnerable Web App)	Web application vulnerability practice	Completed
Host-Only Network Adapter	Isolated lab network (192.168.56.0/24)	Completed

4.3 VM Network Configuration

- Kali Linux IP: 192.168.56.101 (Host-Only Adapter)

- Metasploitable2 IP: 192.168.56.102 (Host-Only Adapter)
- Network Range: 192.168.56.0/24
- Both VMs can communicate with each other but are isolated from the internet
- Connectivity verified using: ping 192.168.56.102 from Kali terminal

5. Linux Fundamentals

Kali Linux command-line proficiency was developed across four key areas:

5.1 File System Navigation

- `cd /path/to/dir` — Change directory
- `ls -la` — List all files including hidden with permissions
- `pwd` — Print current working directory
- `find / -name filename` — Search for files recursively
- `cat`, `less`, `head`, `tail` — View file contents

5.2 File & Directory Permissions

- `chmod 755 file` — Set owner read/write/execute, group/others read/execute
- `chown user:group file` — Change file ownership
- `ls -l` — View permission bits (rwxrwxrwx format)
- `sudo` — Execute commands with elevated privileges
- Understanding SUID/SGID bits and their security implications

5.3 Package Management

- `apt update` & `apt upgrade` — Update package lists and upgrade installed packages
- `apt install package-name` — Install a new package
- `dpkg -i package.deb` — Install a local .deb package
- `apt remove package-name` — Uninstall a package

5.4 Networking Commands

- `ifconfig` / `ip a` — View network interfaces and IP addresses
- `ping 192.168.56.102` — Test connectivity to target
- `netstat -tulnp` — View active listening ports and connections
- `tracert google.com` — Trace the route to a destination
- `ss -tulnp` — Modern replacement for netstat
- `route -n` — View routing table

6. Networking Basics

6.1 OSI Model

The OSI (Open Systems Interconnection) model describes how data travels across a network in 7 distinct layers:

Layer	Name	Function & Security Relevance
7	Application	HTTP, HTTPS, DNS, FTP — target of SQLi, XSS, phishing
6	Presentation	Encryption/decryption, SSL/TLS — data format translation
5	Session	Session management, authentication tokens
4	Transport	TCP/UDP, port numbers — SYN flood attacks target this layer
3	Network	IP addressing, routing — IP spoofing attacks
2	Data Link	MAC addresses, ARP — ARP poisoning/spoofing attacks
1	Physical	Hardware, cables, signals — physical security

6.2 TCP/IP Protocol Suite

- TCP (Transmission Control Protocol): Connection-oriented, reliable delivery with 3-way handshake (SYN → SYN-ACK → ACK). Used by HTTP, HTTPS, SSH.
- UDP (User Datagram Protocol): Connectionless, faster but unreliable. Used by DNS, DHCP, streaming. Exploited in UDP flood attacks.
- IP (Internet Protocol): Handles logical addressing and routing. IPv4 (32-bit) and IPv6 (128-bit).
- ICMP: Used for diagnostics (ping, traceroute). Exploited in ICMP flood/Smurf attacks.

6.3 DNS & HTTP/HTTPS

- DNS (Domain Name System): Resolves domain names to IP addresses. Attacks include DNS spoofing, DNS amplification DDoS, and DNS tunneling.
- HTTP: Stateless application-layer protocol. All data is transmitted in plaintext — vulnerable to MITM attacks.
- HTTPS: HTTP secured with SSL/TLS. Encrypts data in transit. Certificate validation prevents MITM attacks.
- HSTS (HTTP Strict Transport Security): Forces browsers to use HTTPS, preventing SSL stripping attacks.

6.4 IP Addressing, Subnetting & NAT

- IPv4 Classes: Class A (1–126), Class B (128–191), Class C (192–223), Class D (Multicast), Class E (Reserved)
- Private IP Ranges: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16
- Subnetting: /24 = 255.255.255.0 = 254 usable hosts. CIDR notation used to define network size.
- NAT (Network Address Translation): Maps private IPs to a public IP. Makes internal hosts invisible to the internet.
- Lab Network: 192.168.56.0/24 — subnet mask 255.255.255.0, 254 usable addresses

7. Cryptography Basics

7.1 Symmetric vs Asymmetric Encryption

- Symmetric Encryption: Same key used for encryption and decryption. Faster but requires secure key exchange. Examples: AES-256, DES, 3DES. Use case: Encrypting files, disk encryption (BitLocker), VPN tunnels.
- Asymmetric Encryption: Uses a public/private key pair. Public key encrypts, private key decrypts. Slower but solves key distribution problem. Examples: RSA-2048, ECC. Use case: SSL/TLS handshake, digital signatures, PGP email encryption.
- Hybrid Encryption: TLS uses asymmetric encryption to securely exchange a symmetric session key, then switches to symmetric for speed.

7.2 Hashing

- MD5: 128-bit hash. Considered cryptographically broken (collision vulnerabilities). Still used for checksums, not for passwords.
- SHA-1: 160-bit hash. Deprecated for security use due to collision attacks demonstrated in 2017.
- SHA-256: Part of SHA-2 family. 256-bit hash. Currently secure and widely used for file integrity, blockchain, TLS certificates.
- bcrypt/Argon2: Password hashing algorithms with built-in salting and work factor — recommended for password storage.

7.3 Digital Certificates & SSL/TLS

- Digital Certificates: X.509 standard certificates issued by Certificate Authorities (CAs). Bind a public key to an identity.
- SSL/TLS Handshake: Client Hello → Server Hello + Certificate → Key Exchange → Finished. Establishes a secure encrypted channel.
- TLS 1.3: Latest version with improved performance (1-RTT handshake) and removed legacy cipher suites.
- Certificate Chain: Root CA → Intermediate CA → End-Entity Certificate. Browsers validate the chain up to a trusted root CA.

7.4 Hands-On: OpenSSL Encryption

The following OpenSSL commands were practiced for message encryption and decryption:

```
# Generate a 2048-bit RSA private key
openssl genrsa -out private.key 2048
# Extract public key from private key
openssl rsa -in private.key -pubout -out public.key
# Encrypt a message using AES-256-CBC
openssl enc -aes-256-cbc -in message.txt -out message.enc -k secretpassword
# Decrypt the encrypted message
openssl enc -d -aes-256-cbc -in message.enc -out decrypted.txt -k
secretpassword
# Generate SHA-256 hash of a file
openssl dgst -sha256 message.txt
```

8. Tool Familiarization

8.1 Wireshark — Packet Capture & Analysis

- Purpose: Network protocol analyzer for capturing and inspecting packets in real-time.
- Installation: Pre-installed on Kali Linux. Launch with: `wireshark &`
- Capture interface: Selected `eth0` (Host-Only adapter) to capture lab traffic.
- Display filters used: `http, ftp, dns, tcp.flags.syn==1, ip.addr==192.168.56.102`
- Key finding: Captured FTP login — credentials transmitted in plaintext (demonstrated why FTPS/SFTP is necessary).
- Key finding: HTTP requests show full URL paths and headers without encryption.

8.2 Nmap — Network Scanning

- Purpose: Network discovery and security auditing tool.
- Ping scan: `nmap -sn 192.168.56.0/24` — discovered live hosts on the lab network
- TCP SYN scan: `nmap -sS 192.168.56.102` — stealth scan of Metasploitable2
- UDP scan: `nmap -sU 192.168.56.102` — identified UDP services
- Version detection: `nmap -sV 192.168.56.102` — identified service versions (vsftpd 2.3.4, SSH 4.7p1, Apache 2.2.8)
- OS detection: `nmap -O 192.168.56.102` — identified target OS as Linux 2.6.x
- Full scan: `nmap -A 192.168.56.102` — aggressive scan with OS detection + version + scripts

8.3 Burp Suite — Web Application Proxy

- Purpose: Intercepting proxy for testing web application security.
- Configuration: Configured browser proxy to `127.0.0.1:8080` to route traffic through Burp.
- Intercept: Successfully intercepted HTTP requests to DVWA running on Metasploitable2.
- Repeater: Resent modified HTTP requests to test server responses.
- Target Scope: Defined target scope to limit capture to lab IP addresses only.

8.4 Netcat — Network Swiss Army Knife

- Purpose: Versatile networking utility for reading/writing data across TCP/UDP connections.
- Listen mode: `nc -lvp 4444` — set up a listener on port 4444
- Connect mode: `nc 192.168.56.101 4444` — connect to listener from another terminal
- File transfer: Demonstrated transferring a file between VMs using Netcat pipes
- Banner grabbing: `nc 192.168.56.102 22` — grabbed SSH banner to identify service version
- Port scanning: `nc -zv 192.168.56.102 1-1024` — basic port scan using Netcat

9. Deliverables Summary

Component	Details	Status
Lab Setup Report	This document (PDF/DOCX) with screenshots	Completed
GitHub Repository	Notes, cheat-sheets, OpenSSL demo scripts	Completed
Linux Cheat Sheet	Essential commands reference document	Completed
Video Walkthrough	5-min screen recording of lab setup & tools	Completed
Wireshark Capture	Sample pcap file demonstrating packet analysis	Completed
OpenSSL Demo	Encrypt/decrypt script with output screenshots	Completed

10. Key Learnings & Reflections

- Setting up an isolated lab environment is the critical first step in ethical hacking — practicing on live systems without authorization is illegal and unethical.
- The CIA Triad provides a mental model for evaluating every security decision: does this control protect confidentiality, integrity, or availability?
- Linux command-line proficiency is non-negotiable for security professionals — most tools run in terminal and require comfort with the shell.
- Network protocols like FTP and HTTP that transmit data in plaintext are inherently insecure, demonstrating the real-world importance of encryption.
- Nmap is more powerful than most beginners realize — version detection and OS fingerprinting reveal significant information that attackers can exploit.
- Understanding cryptography is fundamental to understanding both attacks (breaking encryption) and defenses (implementing encryption correctly).

11. Next Steps — Task 2 Preview

Having established a solid foundation and working lab environment, Task 2 (Days 13–24) will focus on Network Security & Scanning, including:

- Passive and active reconnaissance techniques (Whois, Google Dorking, Shodan, Nslookup)
- Advanced Nmap scanning with scripting engine (NSE scripts)
- Vulnerability scanning with OpenVAS/Nessus Essentials
- Deep packet analysis with Wireshark — credential extraction from FTP traffic
- SYN flood simulation and analysis using hping3
- Firewall rule creation with iptables

This report was prepared as part of the ApexPlanet Cybersecurity & Ethical Hacking Internship Program.

All activities were performed in an isolated lab environment for educational purposes only.

www.apexplanet.in | info@apexplanet.in | +91 9905879870